A MODULAR-CONTROL REFORMULATION OF THE REGULAR INDUCED SUBGRAPH CONJECTURE

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Abstract. Let $F(n)$ be the largest integer such that every graph on n vertices contains a regular induced subgraph on at least $F(n)$ vertices. We record a structural reformulation of the conjecture that $F(n)$ grows superlogarithmically. For every integer base $b > 1$, the conjecture is equivalent to linear growth along the subsequence b^k, to the corresponding modular target, to a sharper low-degree modular target in which the modulus need only exceed the maximum induced degree, and to an exact one-control formulation with a singleton control vertex. On the modular side, we isolate the one-control modular witness as the canonical asymptotic object and show that the natural multiblock, bucketing, and cascade variants carry no extra asymptotic information: they are all equivalent to the same target. Two transport lemmas and a paired-residue bucketing lemma with quadratic cost in the modulus support this reduction on the finite side. The point is not a new lower bound, but a proof that several apparently stronger modular-control packages collapse to the same frontier.

1. Introduction

All graphs in this paper are finite and simple. For $n \in \mathbb{N}$, let $F(n)$ be the largest integer such that every graph on n vertices contains a regular induced subgraph on at least $F(n)$ vertices. The conjecture usually attributed to Erdős, Fajtlowicz, and Staton asks whether

$$\frac{F(n)}{\log n} \longrightarrow \infty.$$

The basic Ramsey argument gives only the logarithmic lower bound $F(n) \geq c \log n$ for some absolute $c > 0$; compare [5, 1]. Further progress in adjacent settings includes polynomial lower bounds for random graphs [7], algorithmic and structural results for k-regular induced subgraphs [3], and modular-degree theorems in sparse graphs such as the result of Berman, Radcliffe, Scott, Wang, and Wargo that every tree contains a large induced subgraph with all degrees $\equiv 1 \pmod{k}$ [2]. Modular degree conditions also appear in Scott's and Caro's work on induced subgraphs with prescribed residues

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[8, 4]. These results show that congruence constraints arise naturally in induced subgraph problems, but they do not identify the minimal witness package relevant to the regular induced subgraph conjecture. We do not improve any of these bounds.

Instead, the aim of this paper is structural. We isolate a modular-control reformulation of the conjecture in which the remaining open step becomes quite explicit. The contribution is not a new modular-degree theorem, but a normalization result for the conjecture: once one allows ambient-degree and external-degree congruences, there is no asymptotic gain in keeping several control blocks, a dropped part, or a cascade of refinements. Those packages all collapse to a single one-control modular witness target (Theorem 4.4). Concretely, the paper proves three reductions:

- the conjecture is equivalent to its restriction to any geometric subsequence b^k, and to both the usual modular target and the sharper low-degree modular target;
- on the exact side, the same power-scale target is equivalent to the existence of linearly large one-control exact witnesses, even with a singleton control set;
- on the modular side, the one-control, multiblock, bucketing, and cascade witness packages are all asymptotically equivalent.

The key message is therefore not merely that modular language is available, but that the conjecture can be reduced to low-complexity degree data without sacrificing asymptotic strength. The finite lemmas behind this reduction are simple:

- if all vertices of a set S have the same degree in $G[S \cup T]$ and the same degree into a disjoint control set T, then $G[S]$ is regular;
- the modular analogue holds with congruences modulo q;
- when one buckets by the pair consisting of the ambient degree modulo q and the degree into T modulo q, only q^2 residue classes appear.

These lemmas suggest a control-set route that is exponentially cheaper than tracking full adjacency patterns. More importantly, they show that any successful modular-control argument can, after the fact, be normalized to one control set and one modulus. This is the sense in which the reformulation is intended to be canonical rather than cosmetic.

The paper is organized as follows. Section 2 records the geometric-scale and modular reformulations of the conjecture. Section 3 states the finite exact and modular transport lemmas together with the paired-residue bucketing principle. In Section 4 we define the one-control exact and modular witnesses, explain why the exact one-control target is already equivalent to the usual geometric formulation, and isolate the one-control modular target as the canonical modular frontier together with its multiblock, bucketing, and cascade reformulations; Theorem 4.9 then packages the paper as a single equivalence theorem. The final section records a concrete open problem suggested by this reorganization.

Notation. All graphs are finite and simple. For a graph G and a vertex set S, we write $G[S]$ for the induced subgraph on S. For disjoint sets S, T and $v \in S$ we write $\deg_{G[S]}(v)$ for the degree of v in $G[S]$, $\deg_{G[S\cup T]}(v)$ for its degree in $G[S \cup T]$, and $\deg_T(v) = |N_G(v) \cap T|$ for its number of neighbors in T. Congruences $a \equiv b \pmod q$ are taken in $\mathbb{Z}$ in the usual sense; we say a quantity is constant modulo q on a set U if its values on U are pairwise congruent modulo q.

2. Power scales and modular witnesses

Definition 2.1. Fix an integer base $b > 1$.

(1) We say that the *power-sequence target* holds at base b if, for every $M \in \mathbb{N}$, there exists K such that for every $k \geq K$, every graph on b^k vertices contains a regular induced subgraph on at least Mk vertices.
(2) A *modular witness* in a graph G is an induced subgraph $G[S]$ such that all degrees in $G[S]$ are congruent modulo some integer $q \geq |S|$.
(3) A *low-degree modular witness* in a graph G is an induced subgraph $G[S]$ such that all degrees in $G[S]$ are congruent modulo some integer q and every degree in $G[S]$ is strictly smaller than q.
(4) We say that the *power-sequence modular target* holds at base b if, for every $M \in \mathbb{N}$, there exists K such that for every $k \geq K$, every graph on b^k vertices contains a modular witness of order at least Mk.
(5) We say that the *power-sequence low-degree modular target* holds at base b if, for every $M \in \mathbb{N}$, there exists K such that for every $k \geq K$, every graph on b^k vertices contains a low-degree modular witness of order at least Mk.

The first reformulation is the standard but extremely useful observation that the asymptotic statement may be tested on any fixed geometric subsequence.

Theorem 2.2. *For every integer $b > 1$, the following are equivalent.*

(1) $F(n)/\log n \to \infty$.
(2) *The power-sequence target holds at base b.*

Proof. Assume first that $F(n)/\log n \to \infty$. Fix $M \in \mathbb{N}$. Then

$$\frac{F(b^k)}{\log(b^k)} = \frac{F(b^k)}{k \log b} \longrightarrow \infty,$$

so for all sufficiently large k we have $F(b^k) \geq Mk$. This is exactly the power-sequence target.

Conversely, assume the power-sequence target at base b. Let $C > 0$. It is enough to show that $F(n) \geq C \log n$ for all sufficiently large n. Choose an integer $M > 2C \log b$. Let K be such that every graph on b^k vertices contains a regular induced subgraph of order at least Mk for all $k \geq K$. Now let $n \geq b^K$, and choose k with $b^k \leq n < b^{k+1}$.

By monotonicity of F,

$$F(n) \geq F(b^k) \geq Mk.$$

Since $\log n < (k+1)\log b$, we obtain

$$C \log n < C(k+1)\log b \leq \frac{M}{2}(k+1).$$

For all sufficiently large k one has $k+1 \leq 2k$, and then

$$C \log n < Mk \leq F(n).$$

Therefore $F(n)$ eventually dominates every constant multiple of $\log n$, which is equivalent to the conjecture. □

The modular route starts from the observation that congruence plus a sufficiently large modulus is already exact regularity.

Proposition 2.3. *Let G be a graph and let S be a vertex set. If all degrees in $G[S]$ are congruent modulo an integer $q \geq |S|$, then $G[S]$ is regular.*

More generally, if all degrees in $G[S]$ lie in an interval of width strictly smaller than q and are pairwise congruent modulo q, then $G[S]$ is regular.

Proof. Every degree in $G[S]$ lies in $\{0, 1, \ldots, |S|-1\}$. If $q \geq |S|$, then two numbers in that set that are congruent modulo q must coincide. The interval version is the same observation: two integers in an interval of width $< q$ that are congruent modulo q are equal. □

This already converts the geometric regular-subgraph target into a modular one.

Theorem 2.4. *For every integer $b > 1$, the following are equivalent.*

(1) $F(n)/\log n \to \infty$.
(2) *The power-sequence modular target holds at base b.*
(3) *The power-sequence low-degree modular target holds at base b.*

Proof. By Theorem 2.2, it is enough to compare both statements with the power-sequence target. Every regular induced subgraph is trivially a modular witness: take any modulus $q \geq |S|$. It is also a low-degree modular witness: if all degrees in $G[S]$ equal d, take $q = d+1$.

Conversely, Proposition 2.3 turns every modular witness into an exact regular induced subgraph of the same order. A low-degree modular witness is also exact, since all induced degrees lie in $\{0, 1, \ldots, q-1\}$ and are pairwise congruent modulo q. □

Remark 2.5. The modular reformulation is not stronger than the original problem. Its value is organizational: congruence conditions interact better with control sets and bucketing than exact equality does. The low-degree version sharpens this further: the modulus need only dominate the induced degree range, not the full cardinality of the surviving set.

3. Transport lemmas and cheap bucketing

The most basic control-set lemma is the following.

Proposition 3.1 (exact control-set transport)**.** *Let S and T be disjoint vertex sets in a graph G. Suppose that all vertices of S have the same degree in $G[S \cup T]$ and also the same degree into T. Then $G[S]$ is regular.*

Proof. For every $v \in S$,

$$\deg_{G[S]}(v) = \deg_{G[S\cup T]}(v) - \deg_T(v).$$

If the two terms on the right are independent of v, then so is the left-hand side. □

The modular analogue is equally simple.

Proposition 3.2 (modular control-set transport)**.** *Let S and T be disjoint vertex sets in a graph G, and let $q \geq 1$. If the degrees of all vertices of S in $G[S \cup T]$ are pairwise congruent modulo q, and the degrees of all vertices of S into T are pairwise congruent modulo q, then the degrees in $G[S]$ are pairwise congruent modulo q.*

Proof. For $v, w \in S$ one has

$$\deg_{G[S\cup T]}(v) - \deg_{G[S]}(v) = \deg_T(v),$$
$$\deg_{G[S\cup T]}(w) - \deg_{G[S]}(w) = \deg_T(w).$$

Subtracting the two identities and using the congruence hypotheses gives $\deg_{G[S]}(v) \equiv \deg_{G[S]}(w) \pmod{q}$. □

The same bookkeeping works with several control blocks.

Proposition 3.3 (multiblock modular transport)**.** *Let $S, T_1, \ldots, T_r$ be pairwise disjoint vertex sets in a graph G, and let $q \geq 1$. If the degrees of all vertices of S in $G[S \cup T_1 \cup \cdots \cup T_r]$ are pairwise congruent modulo q, and for each $i = 1, \ldots, r$ the degrees of all vertices of S into T_i are pairwise congruent modulo q, then the degrees in $G[S]$ are pairwise congruent modulo q.*

Proof. For every $v \in S$,

$$\deg_{G[S]}(v) = \deg_{G[S\cup T_1\cup\cdots\cup T_r]}(v) - \sum_{i=1}^{r} \deg_{T_i}(v).$$

Each term on the right is constant modulo q on S by hypothesis, so the left-hand side is also constant modulo q on S. □

The key combinatorial gain is that one need only freeze two scalar quantities modulo q, rather than the full adjacency pattern into the control set.

Proposition 3.4 (paired-residue bucketing)**.** *Let S and T be disjoint vertex sets in a graph G, let $q \geq 1$, and let $k \in \mathbb{N}$. If $q^2 k \leq |S|$, then there exists a subset $U \subseteq S$ with $|U| \geq k$ such that all degrees in $G[S]$ on U are congruent modulo q.*

More precisely, one may choose U so that both the ambient degree in $G[S \cup T]$ modulo q and the degree into T modulo q are constant on U.

Proof. Consider the map

$$\phi\colon S \to (\mathbb{Z}/q\mathbb{Z})^2, \qquad \phi(v) = \big(\deg_{G[S\cup T]}(v) \bmod q,\ \deg_T(v) \bmod q\big).$$

Its codomain has exactly q^2 elements. Since $q^2 k \leq |S|$, the pigeonhole principle yields a fiber $U \subseteq S$ with $|U| \geq k$ on which ϕ is constant. Thus all vertices of U have the same degree in $G[S \cup T]$ modulo q and the same degree into T modulo q.

For $u, w \in U$ we have

$$\deg_{G[S]}(u) = \deg_{G[S\cup T]}(u) - \deg_T(u),$$
$$\deg_{G[S]}(w) = \deg_{G[S\cup T]}(w) - \deg_T(w),$$

because S and T are disjoint. The two terms on the right are pairwise congruent modulo q, so $\deg_{G[S]}(u) \equiv \deg_{G[S]}(w) \pmod q$. Hence the degrees in $G[S]$ on U are all congruent modulo q. □

The same pigeonhole argument extends to several control blocks.

Proposition 3.5 (multiblock residue bucketing)**.** *Let $S, T_1, \ldots, T_r$ be pairwise disjoint vertex sets in a graph G, let $q \geq 1$, and let $k \in \mathbb{N}$. If $q^{r+1} k \leq |S|$, then there exists a subset $U \subseteq S$ with $|U| \geq k$ such that all vertices of U have the same degree in $G[S \cup T_1 \cup \cdots \cup T_r]$ modulo q, and for each $i = 1, \ldots, r$ the same degree into T_i modulo q.*

Proof. Consider the map

$$\phi\colon S \to (\mathbb{Z}/q\mathbb{Z})^{r+1},$$
$$\phi(v) = \big(\deg_{G[S\cup T_1\cup\cdots\cup T_r]}(v) \bmod q,\ \deg_{T_1}(v) \bmod q,\ \ldots,\ \deg_{T_r}(v) \bmod q\big).$$

Its codomain has exactly q^{r+1} elements. Since $q^{r+1} k \leq |S|$, the pigeonhole principle yields a fiber $U \subseteq S$ with $|U| \geq k$ on which ϕ is constant. This is exactly the stated residue constancy. □

Thus passing from one control block to r control blocks costs only polynomially many residue classes, namely q^{r+1}, and no additional difficulty in the transport step. The later frontier theorem will show that, asymptotically, this extra bookkeeping does not create a new target.

4. The canonical frontier

We now define the two witness notions that matter most for the asymptotic story.

Definition 4.1. Let G be a graph and $k \in \mathbb{N}$.

(1) A *one-control exact witness of order* k consists of disjoint sets S, T with $T \neq \varnothing$ and $|S| \geq k$ such that all vertices of S have the same degree in $G[S \cup T]$ and the same degree into T.
(2) A *one-control modular witness of order* k consists of disjoint sets S, T with $T \neq \varnothing$, an integer $q \geq |S|$, and $|S| \geq k$ such that all vertices of S have the same degree in $G[S \cup T]$ modulo q and the same degree into T modulo q.

The exact one-control target looks stronger than the power-sequence target, but on geometric scales it is already equivalent to it.

Theorem 4.2 (single-control exact collapse)**.** *Fix an integer $b > 1$. The following are equivalent.*

(1) *The power-sequence target holds at base b.*
(2) *For every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains a one-control exact witness of order Mk.*

In fact, the control set may be chosen to have size 1.

Proof. (2) $\Rightarrow$ (1) follows immediately from Proposition 3.1: every one-control exact witness contains a regular induced subgraph on the surviving set S.

For (1) $\Rightarrow$ (2), assume the power-sequence target at base b. Fix $M \in \mathbb{N}$, and choose K such that every graph on b^k vertices contains a regular induced subgraph on at least $2Mk$ vertices for all $k \geq K$. Let G be a graph on b^k vertices with $k \geq K + 1$, and choose a vertex $t \in V(G)$.

Partition $V(G) \setminus \{t\}$ into the neighbors of t and the non-neighbors of t. One of these two sets has size at least $\lceil (b^k - 1)/2 \rceil$. Since $b > 1$ is an integer, this quantity is at least b^{k-1}. Let W be such a set, and choose a subset $W' \subseteq W$ of cardinality exactly b^{k-1}.

By the power-sequence target applied to the induced subgraph $G[W']$, there exists $S \subseteq W'$ with $|S| \geq 2M(k-1)$ such that $G[S]$ is regular. Every vertex of S has the same degree into the single vertex t, namely either 0 or 1, because all vertices of W lie in the same adjacency class to t. Therefore S together with the control set $\{t\}$ is a one-control exact witness of order $2M(k-1)$ in G.

For $k \geq 2$ one has $2M(k-1) \geq Mk$, so this is in particular a one-control exact witness of order Mk. This proves the required asymptotic statement. □

The modular route admits a larger family of witness notions. We will use the following four variants.

Definition 4.3. Let G be a graph and $k \in \mathbb{N}$.

(1) A *one-control modular bucketing witness of order* k consists of pairwise disjoint sets U, D, T with $T \neq \varnothing$, an integer $q \geq |U|$, and $|U| \geq k$ such that all vertices of U have the same degree in $G[U \cup D \cup T]$

modulo q, the same degree into D modulo q, and the same degree into T modulo q.

(2) A *nonempty modular control-block witness of order* k consists of a set U, a nonempty family of pairwise disjoint control blocks $T_1, \dots, T_r$ disjoint from U, an integer $q \geq |U|$, and $|U| \geq k$ such that all vertices of U have the same degree in $G[U \cup T_1 \cup \cdots \cup T_r]$ modulo q, and for each i they have the same degree into T_i modulo q.

(3) A *modular control-block bucketing witness of order* k consists of a set U, a dropped part D, and a nonempty family of pairwise disjoint control blocks $T_1, \dots, T_r$, all disjoint from one another, together with an integer $q \geq |U|$ and $|U| \geq k$, such that all vertices of U have the same degree in $G[U \cup D \cup T_1 \cup \cdots \cup T_r]$ modulo q, the same degree into D modulo q, and for each i the same degree into T_i modulo q.

(4) A *modular control-block cascade witness of order* k consists of nested sets

$$S_0 \supseteq S_1 \supseteq \cdots \supseteq S_m,$$

a nonempty family of pairwise disjoint control blocks $T_1, \dots, T_r$ disjoint from S_0, and an integer $q \geq |S_m|$, such that $|S_m| \geq k$ and:

(a) for each $i = 1, \dots, m$, all vertices of S_i have the same degree in $G[S_{i-1} \cup T_1 \cup \cdots \cup T_r]$ modulo q and the same degree into the dropped layer $S_{i-1} \setminus S_i$ modulo q;

(b) all vertices of S_m have the same degree in $G[S_m \cup T_1 \cup \cdots \cup T_r]$ modulo q, and for each j they have the same degree into T_j modulo q.

For each of these witness notions, the corresponding *target at base* b means that for every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains such a witness of order Mk.

These variants differ only in whether one explicitly remembers the discarded vertices in a single step or serializes them across a chain of refinements, and in whether the control data are stored in one block or several. The next theorem shows that, asymptotically, they define the same frontier.

Theorem 4.4 (canonical modular frontier)**.** *Fix an integer $b > 1$. The following are equivalent.*

(1) *The one-control modular target holds at base b: for every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains a one-control modular witness of order Mk.*
(2) *The analogous one-control modular bucketing target holds.*
(3) *The analogous nonempty modular control-block target holds.*
(4) *The analogous modular control-block bucketing target holds.*
(5) *The analogous modular control-block cascade target holds.*

Proof. We prove the cycle

$$(1) \Rightarrow (2) \Rightarrow (1), \qquad (1) \Rightarrow (3) \Rightarrow (1),$$
$$(3) \Rightarrow (4) \Rightarrow (3), \qquad (4) \Rightarrow (5) \Rightarrow (4).$$

Together these implications yield the equivalence of all five statements.

(1) $\Rightarrow$ (2): every one-control modular witness is a special case of a one-control modular bucketing witness, namely with empty dropped part $D = \varnothing$.

(2) $\Rightarrow$ (1): let U, D, T be a one-control modular bucketing witness. Since

$$\deg_{G[U\cup T]}(u) = \deg_{G[U\cup D\cup T]}(u) - \deg_D(u)$$

for every $u \in U$, the ambient degrees in $G[U \cup T]$ are constant modulo q on U whenever the ambient degrees in $G[U \cup D \cup T]$ and the degrees into D are constant modulo q on U. The degrees into T are already constant modulo q on U, so U, T, q form a one-control modular witness.

(1) $\Rightarrow$ (3): a one-control modular witness is a nonempty modular control-block witness with exactly one control block.

(3) $\Rightarrow$ (1): let $U, T_1, \dots, T_r$ be a nonempty modular control-block witness, and write $T = T_1 \cup \cdots \cup T_r$. Because the blocks are pairwise disjoint,

$$\deg_T(u) = \deg_{T_1}(u) + \cdots + \deg_{T_r}(u)$$

for every $u \in U$. Since each summand is constant modulo q on U, the total degree into T is also constant modulo q on U. The ambient degree in $G[U\cup T]$ is already constant modulo q, so U, T, q form a one-control modular witness.

(3) $\Rightarrow$ (4): every nonempty modular control-block witness is a modular control-block bucketing witness with empty dropped part.

(4) $\Rightarrow$ (3): let $U, D, T_1, \dots, T_r$ be a modular control-block bucketing witness. For every $u \in U$,

$$\deg_{G[U\cup T_1\cup\cdots\cup T_r]}(u) = \deg_{G[U\cup D\cup T_1\cup\cdots\cup T_r]}(u) - \deg_D(u).$$

Thus the ambient degree on $U\cup T_1\cup\cdots\cup T_r$ is constant modulo q on U whenever the larger ambient degree and the degree into D are constant modulo q on U. The block-degree hypotheses are unchanged, so $U, T_1, \dots, T_r, q$ form a nonempty modular control-block witness.

(4) $\Rightarrow$ (5): let $U, D, T_1, \dots, T_r$ be a modular control-block bucketing witness. Define a one-step chain by

$$S_0 := U \cup D, \qquad S_1 := U.$$

The hypotheses already say that all vertices of S_1 have the same degree in $G[S_0 \cup T_1 \cup \cdots \cup T_r]$ modulo q and the same degree into the dropped layer $S_0 \setminus S_1 = D$ modulo q. To verify the terminal condition, note that for every $u \in U$,

$$\deg_{G[U\cup T_1\cup\cdots\cup T_r]}(u) = \deg_{G[U\cup D\cup T_1\cup\cdots\cup T_r]}(u) - \deg_D(u).$$

Hence the terminal ambient degree is constant modulo q on U, and the block-degree hypotheses are unchanged. Thus $S_0 \supseteq S_1$ together with $T_1, \dots, T_r$ and q is a modular control-block cascade witness.

(5) $\Rightarrow$ (4): let

$$S_0 \supseteq S_1 \supseteq \cdots \supseteq S_m = U$$

be a modular control-block cascade witness with control blocks $T_1, \dots, T_r$ and modulus q. Set

$$D := S_0 \setminus U.$$

If $m = 0$, then $D = \varnothing$, and the terminal condition already gives a modular control-block bucketing witness with empty dropped part.

Assume now that $m \geq 1$. Since $U \subseteq S_1$, the first cascade step implies that all vertices of U have the same degree in $G[S_0 \cup T_1 \cup \cdots \cup T_r]$ modulo q. The terminal condition implies that all vertices of U have the same degree in $G[U \cup T_1 \cup \cdots \cup T_r]$ modulo q. Therefore, for every $u \in U$,

$$\deg_D(u) = \deg_{G[S_0 \cup T_1 \cup \cdots \cup T_r]}(u) - \deg_{G[U \cup T_1 \cup \cdots \cup T_r]}(u)$$

is constant modulo q on U. The terminal block-degree hypotheses are unchanged, and $G[U \cup D \cup T_1 \cup \cdots \cup T_r] = G[S_0 \cup T_1 \cup \cdots \cup T_r]$. Thus $U, D, T_1, \dots, T_r, q$ form a modular control-block bucketing witness. □

The previous two sections combine to show that the modular frontier still implies the original conjecture.

Corollary 4.5. *Fix an integer base $b > 1$. If every sufficiently large graph on b^k vertices contains a one-control modular witness of order Mk for every M, then $F(n)/\log n \to \infty$.*

Proof. By Proposition 3.2, every one-control modular witness yields a modular witness on the surviving set, and Proposition 2.3 turns that modular witness into a regular induced subgraph of the same order. Theorem 2.4 finishes the argument. □

There are also bounded forms worth recording. The first shows that, asymptotically, any eventually positive control budget is as good as a singleton budget on the modular side.

Corollary 4.6 (eventual positive budget)**.** *Fix an integer base $b > 1$, and let $u\colon \mathbb{N} \to \mathbb{N}$ satisfy $u(k) \geq 1$ for all sufficiently large k. Then the following are equivalent:*

1. *For every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains a one-control modular witness of order Mk with a control set of size at most $u(k)$.*
2. *$F(n)/\log n \to \infty$.*

Proof. If (1) holds, then Corollary 4.5 applies a fortiori.

Conversely, assume $F(n)/\log n \to \infty$. Choose K_0 such that $u(k) \geq 1$ for all $k \geq K_0$. By Theorem 4.2, for every $M \in \mathbb{N}$ there exists K_1 such that every graph on b^k vertices with $k \geq K_1$ contains a one-control exact

witness $(S, \{t\})$ of order Mk. For $k \geq \max\{K_0, K_1\}$, the singleton control set respects the budget $u(k)$. Choosing any modulus $q \geq |S|$ turns $(S, \{t\})$ into a one-control modular witness of the same order. □

Corollary 4.7 (singleton budget)**.** *Fix an integer base $b > 1$. The following are equivalent:*

(1) *For every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains a one-control modular witness of order Mk with a control set of size exactly 1.*
(2) $F(n)/\log n \to \infty$.

Proof. This is the special case of Corollary 4.6 with $u(k) = 1$. □

Corollary 4.8 (sublinear budgets collapse modular to exact)**.** *Fix an integer base $b > 1$, and let $u\colon \mathbb{N} \to \mathbb{N}$ satisfy $u(k) = o(k)$ and $u(k) \geq 1$ for all sufficiently large k. Then the following are equivalent:*

(1) *For every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains a one-control modular witness of order Mk with a control set of size at most $u(k)$.*
(2) *For every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains a one-control exact witness of order Mk with a control set of size at most $u(k)$.*
(3) $F(n)/\log n \to \infty$.

Proof. Corollary 4.6 gives the equivalence of (1) and (3). Also, (3) implies (2) by Theorem 4.2, because the singleton control set is eventually allowed by the assumption $u(k) \geq 1$.

It remains to prove (1)⇒(2). Fix $M \in \mathbb{N}$. Because $u(k) = o(k)$, there exists K_0 such that $u(k) < Mk$ for all $k \geq K_0$. By (1), there exists K_1 such that every graph on b^k vertices with $k \geq K_1$ contains a one-control modular witness (S, T, q) of order Mk with $|T| \leq u(k)$. For $k \geq \max\{K_0, K_1\}$, we therefore have

$$|T| \leq u(k) < Mk \leq |S| \leq q.$$

Since all degrees into T are congruent modulo q and lie in the interval $\{0, 1, \ldots, |T|\}$ of width $< q$, they are all equal; write this common value as e. By Proposition 3.2, the degrees in $G[S]$ are pairwise congruent modulo q, and Proposition 2.3 then shows that $G[S]$ is regular because $q \geq |S|$. Let d be the common degree in $G[S]$. Then every vertex of S has degree $d + e$ in $G[S \cup T]$, so (S, T) is a one-control exact witness of order Mk with control set size at most $u(k)$.

Finally, (2)⇒(1) is immediate by choosing any modulus $q \geq |S|$. □

Theorem 4.9 (summary reformulation package)**.** *Fix an integer base $b > 1$. The following are equivalent.*

(1) $F(n)/\log n \to \infty$.
(2) *The power-sequence target holds at base b.*
(3) *The power-sequence modular target holds at base b.*

(4) *The power-sequence low-degree modular target holds at base b.*
(5) *For every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains a one-control exact witness of order Mk.*
(6) *The one-control modular target holds at base b.*
(7) *The analogous one-control modular bucketing target holds.*
(8) *The analogous nonempty modular control-block target holds.*
(9) *The analogous modular control-block bucketing target holds.*
(10) *The analogous modular control-block cascade target holds.*
(11) *For every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains a one-control modular witness of order Mk with a control set of size exactly* 1.

Proof. The equivalence of (1), (2), (3), and (4) is Theorems 2.2 and 2.4. The equivalence of (2) and (5) is Theorem 4.2. Statements (6)–(10) are equivalent by Theorem 4.4, and (6) implies (1) by Corollary 4.5. Finally, (1) and (11) are equivalent by Corollary 4.7, while (5) trivially implies (6) by choosing any modulus $q \geq |S|$. Together these implications identify all eleven statements. □

Remark 4.10. Theorem 4.9 is the sense in which the paper is intended as a genuine reduction rather than a vocabulary change. It shows that several witness frameworks that might naturally be pursued as separate programs — larger control families, explicit dropped parts, or cascades of refinements — do not define distinct asymptotic problems. Any proof in one of those settings can be normalized to the one-control modular target without asymptotic loss. Accordingly, the more elaborate witness packages can matter only if they retain additional quantitative information beyond the bare existence statements considered here. The mathematical content of the paper is therefore the removal of these false frontiers.

5. Concrete open problems

The reformulation above leaves one canonical equivalent problem, one stronger exact route, and one conditional dyadic-lifting criterion.

Problem 5.1 (canonical modular frontier)**.** Fix an integer base $b > 1$. Is it true that for every $M \in \mathbb{N}$, every sufficiently large graph on b^k vertices contains a one-control modular witness of order Mk?

By Theorem 4.9, Problem 5.1 is equivalent to the regular induced subgraph conjecture. It is the normalized frontier isolated by the paper: any existence proof that proceeds through multiblock, bucketing, or cascade witnesses can be converted back to this one-control target without asymptotic loss.

A stronger route would be to force exact control data with sublinear control size.

Problem 5.2 (sublinear exact control)**.** Is there a function $u(m) = o(m)$ such that every sufficiently large graph contains disjoint sets S, T with

$$|S| \geq m, \qquad |T| \leq u(m),$$

all vertices of S having the same degree in $G[S \cup T]$, and all vertices of S having the same degree into T?

By Proposition 3.1, a positive answer to Problem 5.2 would imply that $F(n) \to \infty$: for each fixed m, every sufficiently large graph would then contain a regular induced subgraph on at least m vertices. That qualitative conclusion alone does not recover the conjectured superlogarithmic lower bound, because the threshold beyond which the witness appears may depend arbitrarily on m. Thus this stronger problem is not claimed equivalent to the conjecture; it is included because it isolates one natural exact-control route beyond the canonical modular frontier. The modular route suggests an intermediate target: replace exact equalities by congruences modulo a modulus q comparable to $|S|$, or even more sharply by congruences modulo a modulus larger than the induced degree range, and then appeal to Proposition 2.3. In either form, the obstacle is no longer definitional. The canonical open step is Problem 5.1; Problem 5.2 asks whether one can do better by forcing a more rigid witness.

A dyadic modulus-lifting route. The canonical frontier also suggests a quantitative modulus-by-modulus program. Let $\mathcal{W}_{2^j}$ be any family of witness notions indexed by powers of two, where a witness in $\mathcal{W}_{2^j}$ comes with a terminal vertex set on which all induced degrees are congruent modulo 2^j.

Proposition 5.3 (dyadic lifting criterion)**.** *Assume that the family $(\mathcal{W}_{2^j})_{j\geq 1}$ has the following properties.*

1. *Every graph on n vertices has a witness in $\mathcal{W}_2$ of order at least $n/2$.*
2. *There is an absolute constant C such that for every $j \geq 1$ and every $m \in \mathbb{N}$, every graph with a witness in $\mathcal{W}_{2^j}$ of order at least $(2^j)^C m$ also has a witness in $\mathcal{W}_{2^{j+1}}$ of order at least m.*
3. *Every witness in $\mathcal{W}_{2^j}$ of order at least 2^j yields a regular induced subgraph on 2^j vertices.*

Then $F(n) \geq \exp(\Omega(\sqrt{\log n}))$; in particular $F(n)/\log n \to \infty$.

Proof. For a power of two q, let $W_q(n)$ denote the largest integer m such that every graph on n vertices has a witness in $\mathcal{W}_q$ of order at least m. By (1), $W_2(n) \geq n/2$. Iterating (2) from 2 to 2^r gives

$$W_{2^r}(n) \geq \left\lfloor \frac{n}{2\prod_{j=1}^{r-1}(2^j)^C} \right\rfloor = \left\lfloor \frac{n}{2^{1+Cr(r-1)/2}} \right\rfloor.$$

Choose r maximal with

$$1 + r + \frac{Cr(r-1)}{2} \leq \log_2 n.$$

Then

$$\frac{n}{2^{1+Cr(r-1)/2}} \geq 2^r,$$

so in particular $2^r \leq W_{2^r}(n)$. By (3), every graph on n vertices therefore contains a regular induced subgraph on 2^r vertices. Since $r = \Omega(\sqrt{\log n})$, we obtain

$$F(n) \geq 2^r = \exp(\Omega(\sqrt{\log n})),$$

and in particular $F(n)/\log n \to \infty$. □

Gallai's classical parity theorem gives the base case at modulus 2 for the empty-control package of induced subgraphs with all degrees even [6]. The remaining difficulty is therefore not the first modulus, but finding a composable fixed-modulus witness notion for which the lift in Proposition 5.3(2) can actually be proved and for which the collapse in (3) is available at the terminal scale.

To summarize: the conjecture is equivalent to linear growth along any geometric subsequence, to the corresponding modular target, to the sharper low-degree modular target, to the exact one-control target with a singleton control vertex, and to each of the modular-control packages listed in Theorem 4.9. Problem 5.1 records the normalized equivalent frontier, while Problem 5.2 records a stronger exact route. The finite lemmas that support this viewpoint replace full neighborhood data by ambient and external degree data, and replace exponentially many profile classes by polynomially many residue classes. Proposition 5.3 records a separate conditional route: a polynomial-cost dyadic lift in any composable fixed-modulus package would force the quantitative bound $F(n) \geq \exp(\Omega(\sqrt{\log n}))$, which comfortably dominates the conjectured superlogarithmic growth. That is the structural contribution of the present paper.

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